OMNI UGV / REV A PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

Battery protection

File: battery_protection.kicad_sch

File: interlock.kicad_sch

1S LiPo / ESP32-S3 / three brushed motors / OV2640

Verify motor terminal fit and assembled power integrity before fabrication release.

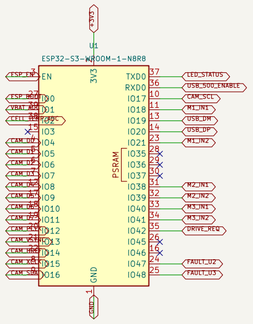
Nathaniel Sanford

Sheet: /
File: omni_ugv.kicad_sch

Title: Omni UGV / Rev A

Size: A3	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 1/7

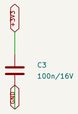
CONTROLLER / NET-LABEL CONNECTIONS











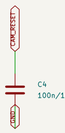


































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Sheet: /1. Controller/
File: controller.kicad_sch

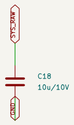
Title: Omni UGV / Controller

Size: A2	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 2/7

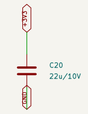
REGULATOR / NET-LABEL CONNECTIONS



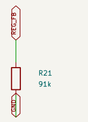


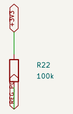


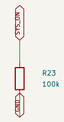


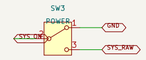






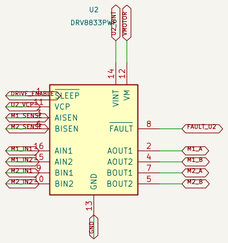


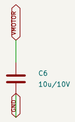


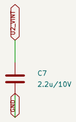


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Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: Omni UGV / Regulator		
Size: A2	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 3/7

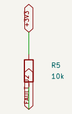
MOTORS / NET-LABEL CONNECTIONS

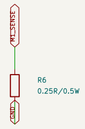


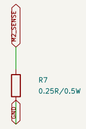


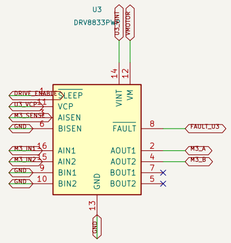


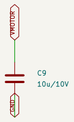


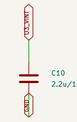




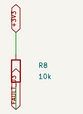


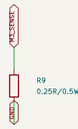




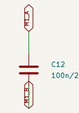




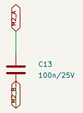




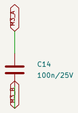


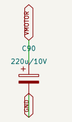












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Sheet: /2. Motors/
File: motors.kicad_sch

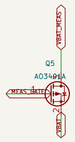
Title: Omni UGV / Motors

Size: A2	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 4/7

INTERLOCK / NET-LABEL CONNECTIONS







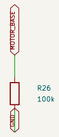












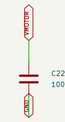




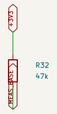


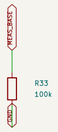




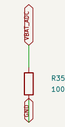


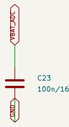












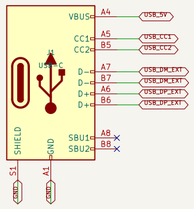
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Sheet: /5. Interlock/
File: interlock.kicad_sch

Title: Omni UGV / Interlock

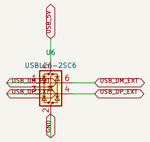
Size: A2	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 5/7

CHARGING / NET-LABEL CONNECTIONS

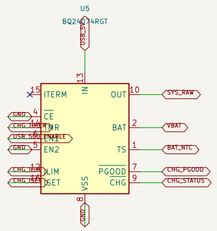


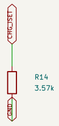


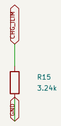






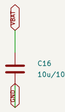
























J2 is RAW CELL: no ground connection except through R36. J4 uses an electrically isolated two-wire sensor bonded to the CELL.

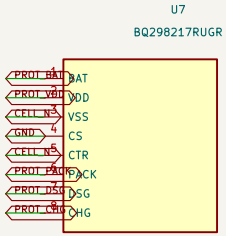
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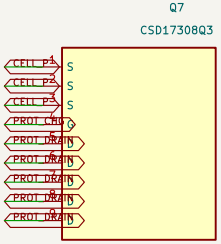
Sheet: /3. Charging/
File: charging.kicad_sch

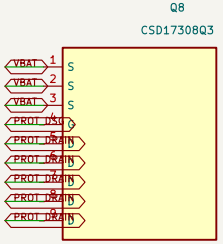
Title: Omni UGV / Charging

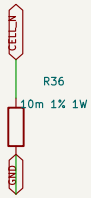
Size: A2	Date: 2026-11-02	**Rev: A-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 6/7

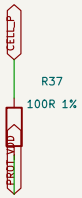
RAW CELL: J2 -> Q7/Q8 -> VBAT (charger + motors). R36 is the ONLY CELL_N-to-GND connection.
U7 CS and VSS require Kelvin routing to opposite R36 pads. No bypass via sensor wires or USB.

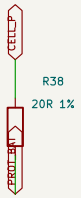


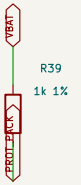


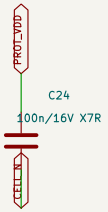


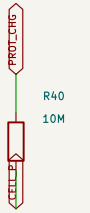


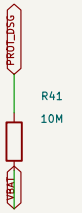






















J4: independent charger probe. J5: second 103AT-2 cell probe for GPIO1/ADC1.
Both probes contact the cell pouch through electrical insulation; neither mounts on this PCB.
Open/short J5 must disable driving. Charge temperature control remains hardware BQ24074 TS.

Sheet: /Battery protection/
File: battery_protection.kicad_sch

Title: 1S protection and CELL temperature

Size: A4	Date:	Rev: B-PROTOTYPE
KiCad E.D.A. 10.0.8		Id: 7/7